

Information Security

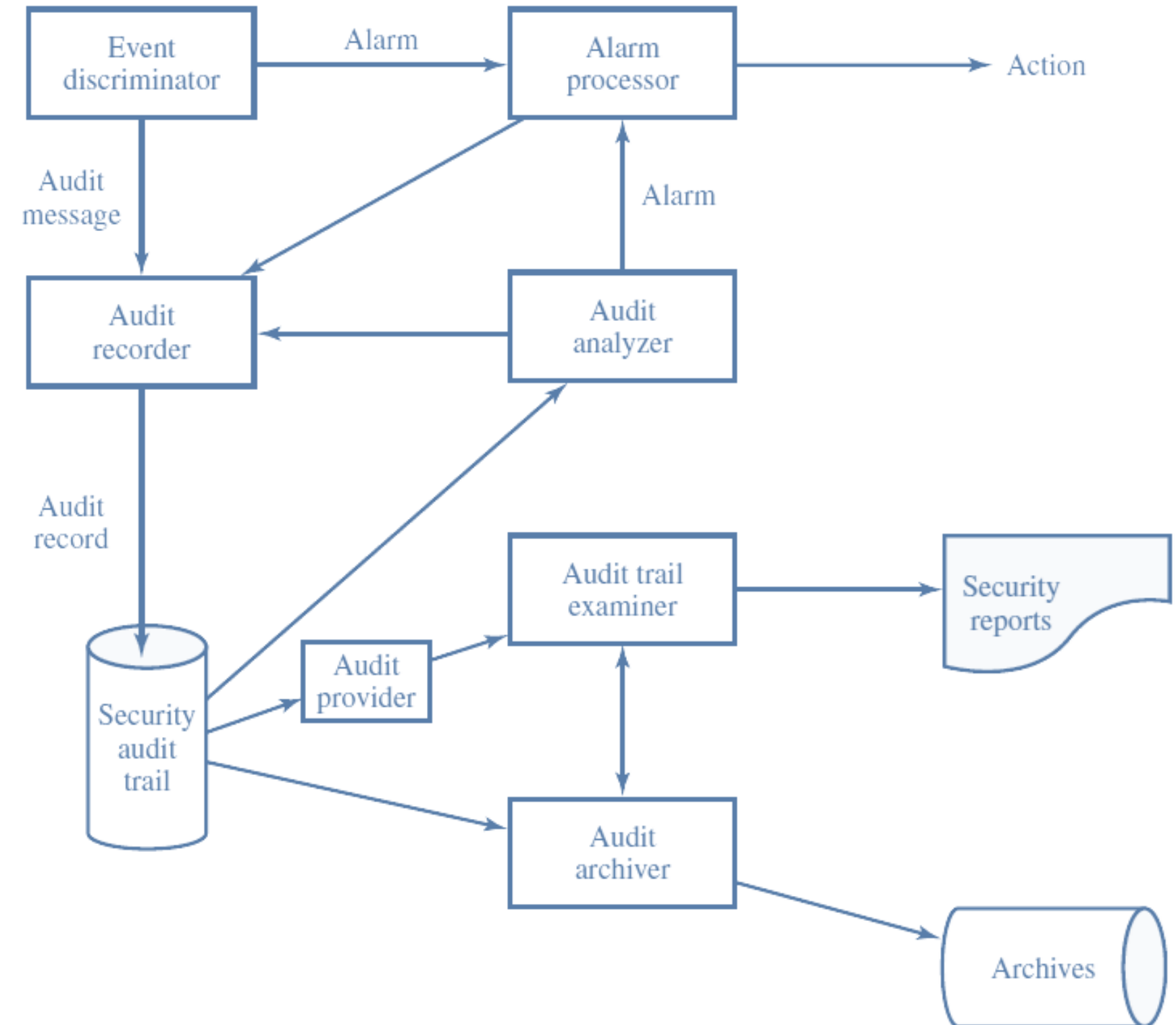
Unit- 8 Security Audit

Security Audit

- A security audit is an independent review and examination of an organization's information system (IS) records and activities to ensure proper security measures are in place. It helps verify whether the system complies with established security policies and procedures, detects security breaches, and suggests countermeasures if needed. The main goal is to hold entities accountable for their actions in security-related events and ensure system integrity.
- Key benefits of a security audit include:
 1. Assurance of system security operations.
 2. Data generation for post-attack analysis.
 3. Identification of security gaps.
 4. Detection of abnormal behavior.
 5. Collection of evidence for computer forensics.
- Security audits rely on audit trails (also known as audit logs), which are chronological records that document system activities. These records help reconstruct and analyze events in a security-relevant transaction, assisting in detecting and investigating potential security breaches or attacks.

Security Auditing Architecture

- Security auditing architecture is a structured approach to monitor, record, and analyze the activities within an information system to ensure that everything is secure.
- This architecture consists of several components working together to identify and respond to security-related events.



1. Event Discriminator :

- This is like a security guard inside the system. It is a software that constantly watches the system to spot any suspicious or important events (like someone trying to access secure data). It is configured to detect specific security-related activities.

2. Audit Recorder :

- Once the event discriminator (our security guard) notices something, it sends this information to the audit recorder. The audit recorder then logs this event into a record. This record can be thought of as a detailed note of what happened.

3. Alarm Processor :

- Some events are more serious and need immediate attention. These are called alarm events. When the event discriminator detects one, it sends an alarm to the alarm processor, which takes necessary action (like alerting a human operator or blocking access). This action is also recorded by the audit recorder.

4. Security Audit Trail :

- The audit recorder keeps all the recorded events in a log known as the security audit trail. This is like a diary that keeps track of everything that happens, step by step.

5. Audit Analyzer :

- The audit analyzer looks at the security audit trail to spot patterns or trends that might indicate a security problem. For example, if it notices repeated failed login attempts, it might raise an alarm or create a new event to be recorded.

6. Audit Archiver :

- Over time, the audit trail can get quite large. The audit archiver helps manage this by moving older records into long-term storage, creating an archive. This archive can be used later for historical analysis or investigations.

7. Audit Provider :

- This is the interface or application that allows people (like security professionals) to access and use the audit trail data. It makes it easier to find and review specific events.

8. Audit Trail Examiner :

- The audit trail examiner is the person or software that reviews the audit trail and archives. This role is crucial for understanding past events, identifying trends, or gathering evidence for investigations (like in computer forensics).

9. Security Reports :

- After examining the audit trail, the examiner generates security reports. These reports summarize the findings in a way that humans can easily understand. They help in making informed decisions about the security of the system.

Security Audit Trail

- A security audit trail is a detailed record or log that keeps track of everything happening in a computer system to help detect and analyze any security issues.
- It is like a digital library that notes down every important event, especially those related to security.

What to collect :

- When setting up a security audit trail, you have to decide what information to collect.
- The choice depends on how much detail you want and how much data the system can handle without slowing down.
- Collecting too much information can overwhelm the system and make it harder to find important details.
- Here are few thing that might be recorded:

1. Event Related to Security :

- This includes everything related to security, such as attempts to log in, use of security features, and any alarms triggered by the system.

2. System Operations :

- This might include actions like starting or stopping the system, connecting or disconnecting devices, and any changes made to system settings.

3. Access Information :

- Who accessed the system, when they did it, and what they try to do.

4. Remote Access :

- Any attempts to access the system from outside, like from another computer.

Why Collect Data :

- The main goal of collecting this data is to help to identify normal behavior versus abnormal behavior, which could indicate a security threat. It also helps in understanding what happened during an attack and finding ways to prevent future ones.

Types of Audit Trails:

1. System-Level Audit Trails :

- These are used to monitor the overall system, like tracking login attempts, device usage, and other system operation.

2. Application –Level Audit Trails :

- These focus on specific applications, like recording who sent and received emails or what data was accessed in a database.

3. User–Level Audit Trails :

- These track the actions of individual users, such as commands they issued and files they accessed.

4. Physical Access Audit Trails :

- These come from devices that control physical access to places, like card-key systems. They record who tried to enter a building or room and whether they were successful.

Protecting the Audit Trails:

- It is important to protect these records because they contain sensitive information. Here are some ways to do that:

1. Store Securely :

- Use methods like write-once media (CD-ROMs) or print logs that can't be easily altered.

2. Integrity and Confidentiality :

- Ensure that the data hasn't been tampered with and that only authorized people can view it. This can be done by using digital signatures and encryption.

Audit Trail Analysis

- Audit Trail Analysis involves examining records of activities (logs) on a system to monitor security and understand user behavior.

1. Understanding Logs :

- Logs are the digital footprints lefts by users and systems. They record actions like logging in, accessing files, or making changes.
- To make sense of these logs, it is important to understand what's normal and what's not. This might require looking at other logs or knowing the usual behavior of users and systems.

2. Timing Analysis :

- If something unusual happens, like a security breach or system error, looking at the logs helps to figure out what went wrong and how to fix it.
- Regularly checking logs helps spot problems before they become serious. It also helps in understanding typical system behavior, making it easier to spot anything unusual.

3. Audit Review :

- Sometimes, we only need to look at specific logs, like those related to particular user or event. Tools can help by filtering out unnecessary information and highlighting important details.

!!!Thank you!!!